Management of TPA-Induced Angioedema

Overview

Angioedema is a rare but potentially life-threatening complication of tissue plasminogen activator administration, occurring in up to **1–5% of patients**, especially in those with **middle cerebral artery (MCA) infarcts** or **ACE inhibitor use**. Believed to be **bradykinin-mediated**, not histamine-driven (**ACE inhibitors**, which increase bradykinin, can raise the risk).

Recognition

Time of Onset:

- Typically occurs within **1–2 hours** after tPA administration, often during or shortly after the infusion.

Presentation:

- Unilateral or bilateral **facial swelling**
- **Tongue, lips, or airway** involvement
- May be **asymmetric**, often on the side **contralateral** to the stroke
- May or may not have urticaria or rash

Early recognition is critical to prevent **airway compromise**.

Risk Factors

- **Use of ACE inhibitors**
- **Large infarct involving the insular cortex or MCA territory**
- **Female sex**
- **History of angioedema**

Management Protocol

◇ Step 1: Stop the Infusion

- Immediately **discontinue TPA infusion** (TNK a bolus so this would only apply to alteplase if still infusing**)** if angioedema is suspected or observed.

◇ Step 2: Secure the Airway

- Assess for signs of airway obstruction
- Call anesthesia or ENT early if airway involvement is suspected
- Prepare for **emergent intubation** if progressive swelling or stridor

◇ Step 3: Administer Medications

- **IV Corticosteroids:**
  - **Methylprednisolone 125 mg IV** once
- **IV Antihistamines:**
  - **Diphenhydramine 50 mg IV**
  - **Famotidine 20 mg IV** (H2 blocker)
- **Epinephrine (if systemic symptoms or airway involvement):**
  - **0.3–0.5 mg IM** (1:1000) every 5–15 minutes as needed
  - Or **0.1 mg IV push** (1:10,000) for hypotension or airway compromise

Step 4: Supportive Care & Monitoring

- Close monitoring in the **ICU or ED resuscitation area**
- Frequent reassessment of swelling and respiratory status
- Consider **C1 esterase inhibitor (Berinert)** or **Icatibant (Firazyr)** for refractory cases (based on institutional availability and allergist consultation)
- Tranexamic acid (TXA) has been used in cases of ACE inhibitor–induced angioedema; however, its use in tPA-induced angioedema should be approached with caution. In patients with acute ischemic stroke (AIS), the potential benefit must be carefully weighed against the increased risk of thromboembolic complications, given the prothrombotic state and recent administration of a thrombolytic agent.

Documentation & Reporting

- Document time of tPA administration, time of onset, symptoms, and treatments given
- Report to institutional safety reporting systems and stroke registry, if applicable

References

1. Hill MD, Lye T, Moss H, et al. Hemi-orolingual angioedema and ACE inhibition after alteplase treatment of stroke. *Neurology*. 2003;60(9):1525–1527. doi:10.1212/01.WNL.0000063315.36590.EE
2. Suri MF, Nasar A, Qureshi AI. Prevalence and outcomes of orolingual angioedema in acute ischemic stroke patients treated with thrombolysis. *Stroke*. 2006;37(12):e155–e157. doi:10.1161/01.STR.0000252133.15476.54
3. Powers WJ, Rabinstein AA, Ackerson T, et al. 2019 Guidelines for the Early Management of Acute Ischemic Stroke. *Stroke*. 2019;50(12):e344–e418. doi:10.1161/STR.0000000000000211